

Elijah Pivo

Curriculum Vitae

MIT Sloan School of Management
Operations Research and Statistics
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EDUCATION

- 2026 PhD, Social and Engineering Systems
Massachusetts Institute of Technology
Institute for Data, Systems, and Society
Dissertation: *A Holistic Approach for the Simulation, Evaluation, and Design of Organ Transplantation Policies in the U.S.*
Advisor: Dimitris Bertsimas
- 2018 BS, Electrical Engineering and Computer Engineering
Johns Hopkins University

PROFESSIONAL APPOINTMENTS

- 2026 – Present Postdoctoral Associate
MIT Sloan School of Management
- 2018 – 2019 Robotics Engineer
Honeybee Robotics

PUBLICATIONS

Refereed Journal Articles

- 2025 **Elijah Pivo**, James Alcorn, Dimitris Bertsimas, Sarah E Booker, Keighly Bradbrook, Thomas G Dolan, Lindsay V Larin, Kayla R Temple, and Nikolaos Trichakis. “Modernizing the Design Process for US Organ Allocation Policy: Toward a Continuous Distribution Policy for Kidneys.” *Transplantation Direct*, 11(9), e1780.
- 2022 Richard Pates, Andres Ferragut, **Elijah Pivo**, Pencheng You, Fernando Paganini, and Enrique Mallada. “Respect the Unstable: Delays and Saturation in Contact Tracing for Disease Control.” *SIAM Journal on Control and Optimization*, 60(2), S196–S220.

Manuscripts in Preparation

- 2026 **Elijah Pivo**, Thomas Athey, Nikolaos Trichakis, and Dimitris Bertsimas. "Addressing Geographic Access Disparities Under Time-Sensitive Logistics in Kidney Allocation Policy." In preparation for *Management Science*.
- 2026 Thomas Athey, **Elijah Pivo**, Matthew Hartwig, Marie Budev, Dennis Lyu, Kaitlin Swanner, Keighly Bradbrook, Nikolaos Trichakis, and Dimitris Bertsimas. "Optimization-Based Design of Organ Allocation Policy: Mitigating Biologic Access Disparities in Lung Transplantation." In preparation for *American Journal of Transplantation*.

Grants in Preparation

- 2026 "NSF TTP-P: A National Platform for Simulation-Driven Design of U.S. Organ Allocation Policy." In preparation for submission to the NSF Translation to Practice program. In partnership with UNOS; collaborators include Dimitris Bertsimas and Kaitlin Swanner.

Published Conference Abstracts and Conference Proceedings

- 2026 **Elijah Pivo**, Thomas Athey, Maura Hegarty, Nikolaos Trichakis, and Dimitris Bertsimas. "A Unified, Fast, and Transparent System for Designing Organ Allocation Policies." *American Journal of Transplantation* 26(7), S257. Abstract Only.
- 2026 Thomas Athey, **Elijah Pivo**, and Dimitris Bertsimas. "Lung Allocation Policy Optimization to Equalize Transplant Access Across Blood Types." *American Journal of Transplantation*, 26(7), S1028. Abstract Only.
- 2023 **Elijah Pivo**, Nikolaos Trichakis, Dimitris Bertsimas, Sarah Booker, Keighly Bradbrook, Joanne White, Lindsay Larkin, Thomas Dolan, and James Alcorn. "Powerful New Methods and Tools for Designing Allocation Policy for Deceased-Donor Organs." *American Journal of Transplantation*, 23(6), S474-S474. Abstract Only.
- 2016 Chris Assad, Javier Rodriguez, **Elijah Pivo**, Calvin Huang, Michael Pomerantz, and Adrian Stoica. "Live Demonstration: BioSleeve, a Wearable Hands-Free Gesture Control Interface." *IEEE Sensors*, Orlando, FL, USA, 30 Oct–3 Nov.

AWARDS AND HONORS

- 2018 William H. Huggins Award, Electrical and Computer Engineering, Johns Hopkins University
- 2017 Tau Beta Pi Engineering Honor Society, Elected to Membership as a Junior

GRANTS AND FELLOWSHIPS

- 2025 MIT HEALS Graduate Fellowship
- 2020 U.S. National Science Foundation Graduate Research Fellowship Program
- 2020 MIT IDSS Michael Hammer Fellowship

INVITED TALKS

- 2024 “AI and Policy Development,” *The Organ Donation and Transplantation Alliance National Critical Issues Forum*, Orlando, FL, 6 Sep.

CONFERENCE PARTICIPATION

Sessions Organized

- 2026 “Optimization for Societal Systems,” *INFORMS Optimization Society Conference*, Atlanta, GA, 20–22 Mar.

Presentations

- 2026 “Advancing the Simulation, Evaluation, and Design of Organ Transplantation Policies in the U.S.,” *INFORMS Annual Meeting*, San Francisco, CA, 1 Nov.
- 2026 “Modernizing the Development of Organ Allocation Policy,” *INFORMS Healthcare Conference*, Raleigh, NC, 28 Jul.
- 2026 “A Unified, Fast, and Transparent System for Designing Organ Allocation Policies,” *American Transplant Congress*, Boston, MA, 22 Jun.
- 2026 “Advancing the Design of U.S. Organ Transplant Policy with Optimization,” *INFORMS Optimization Society Conference*, Atlanta, GA, 21 Mar.
- 2025 “Solving for Welfare, Fairness, and Feasibility: Modern Computational Methods for the U.S. Organ Transplant System,” *INFORMS Annual Meeting*, Atlanta, GA, 26 Oct.
- 2024 “Thousands Unused: The Problem of Lossy Allocation in the National Kidney Transplant System and Opportunities for Operations Research,” *INFORMS Annual Meeting*, Seattle, WA, 23 Oct.
- 2023 “Powerful New Methods and Tools for Designing Allocation Policy for Deceased-Donor Organs,” *American Transplant Congress*, San Diego, CA, 5 Jun.
- 2023 “Bringing Modern Analytics to Kidney Transplantation Policy Design,” *INFORMS Annual Meeting*, Phoenix, AZ, 15 Oct.
- 2022 “An Inverse Optimization Approach to Causal Inference,” *INFORMS Annual Meeting*, Indianapolis, IN, Analytics and AI, 16 Oct.

CAMPUS AND DEPARTMENTAL TALKS

- 2026 Panelist, "Data-Driven Decision Making," *SES10: 10th Anniversary Symposium of the SES Doctoral Program*, Cambridge, MA, 9 Oct.
- 2026 "Operations Research for the U.S. Organ Transplant System," *MIT HEALS Annual Symposium*, Cambridge, MA, 16 Mar.
- 2026 "Operations Research for the U.S. Organ Transplant System: Kidney Transportation Logistics," *Holistic AI for Medicine*, Cambridge, MA, 25 Feb.

TEACHING EXPERIENCE

Massachusetts Institute of Technology

Optimization Methods (Fall 2024)

Analytics for a Better World (Spring 2022)

Johns Hopkins University

Control Systems (Spring 2020)

EXTRA TRAINING

- 2024 Massachusetts Institute of Technology Kaufman Teaching Certificate

COMMUNITY INVOLVEMENT AND OUTREACH

- 2020 STEM Achievement in Baltimore Elementary Schools Instructor

RELATED PROFESSIONAL SKILLS

Programming & Computing: Julia, Python, C/C++, Java, MATLAB, Git, Linux, Slurm

Optimization: JuMP, Gurobi, MOSEK, Knitro

Web & Databases: Vue.js, AWS, SQL, SQLite

Embedded Systems: PCB Design & Fabrication (KiCad), FPGA Programming (VHDL), Microcontrollers

Engineering & Robotics: SolidWorks, SysML, ROS

PROFESSIONAL MEMBERSHIPS

Institute for Operations Research and the Management Sciences (INFORMS),
Member

REFERENCES

Dimitris Bertsimas, PhD

Boeing Leaders for Global Operations Professor of Management
Professor of Operations Research
MIT Sloan School of Management
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Nikolaos Trichakis, PhD

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Kaitlin Swanner, MPP

Senior Strategic Initiatives Manager
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